What next?

US president George W. Bush pitched high objectives in 2004 when he stated that "Humans are headed into the cosmos." Certainly, a return to the Moon seems likely and it is even feasible that humans could set foot on Mars sometime this century. But beyond that, informed speculation stumbles.

Some visionaries have long predicted a mass migration of humans to other worlds. In 1926 Konstantin Tsiolkovsky, the Russian prophet of space travel, had a vision of the colonization of planets on a massive scale, with "the population of the solar system... one hundred thousand million times the current population of the Earth." American rocket pioneer Robert Goddard believed that, when the Sun died, human life could continue through a "last migration" of interstellar "arks" in search of another inhabitable planet. More recently, physicist Stephen Hawking predicted the need for such migration much sooner, asserting that "The human race will not survive the next thousand years unless we spread into space."

Against such visions stand what is known about the impossibly long time required for even the fastest conceivable journey to any other star and the implacable hostility to human life of the environments found everywhere in our Solar System except on planet Earth. One of the most consistent experiences of those who have spent long periods in space has been nostalgia for the simple smells, sights, and sounds of nature. As Russian cosmonaut Vladimir Lyakhov said with telling simplicity, Earth is the place "where we will always strive to return. It is our home."

Remote warfare

The way in which aerial warfare has developed in the 21st century would once have sounded as surreal as interstellar colonization. The notion of a fleet of remote-controlled, pilotless aircraft patrolling the planet, ready to assassinate a suspected enemy of the United States anywhere in the world at the press of a button, would have seemed to belong to the realm of fantasy fiction. Yet Unmanned (robotic) Aerial Vehicles (UAVs), also known as drones, have become a ubiquitous reality of modern air war, and have many civilian applications. The first military use of UAVs was primarily for intelligence and reconnaissance. The Northrop Grumman Global Hawk, for example, proved an excellent platform for high-altitude surveillance equipment, fulfilling the role once performed by manned aircraft such as the SR-71 and the U-2. After the invasion of Afghanistan in 2001, the United States discovered how effective the Predator UAV could be as a hunter-killer when armed with Hellfire missiles. The introduction of the grimly named Reaper in the same attack role followed in 2007. UAVs were in many ways the ideal vehicles for the kind of wars the United States found

EUROFIGHTER

The Eurofighter Typhoon is a multirole fighter that first saw combat in the military intervention in Libya in 2011. It can cruise at supersonic speed without afterburning and achieves outstanding agility through an inherently unstable design and lightweight construction.

SOLO AROUND THE WORLD

Over 16 months from March 2015 to July 2016, Swiss flight enthusiasts André Borschberg and Bertrand Piccard piloted their aircraft Solar Impulse 2 on the first solar-powered circumnavigation of the globe. Heading eastward from Abu Dhabi in the United Arab Emirates, they reached East Asia by the end of May. The most demanding leg of the journey, across the Pacific from Japan to Hawaii, took almost five days to complete at a speed of 30–60mph (50–100kph), was the longest solo flight ever made. Technical problems and adverse weather delayed resumption of the flight until April 2016, but the aircraft reached New York in June and then made a three-day crossing of the Atlantic to Seville, Spain. The journey on to Abu Dhabi was completed in two stages, with a stop in Cairo, Egypt. Borschberg and Piccard felt they had proved the effectiveness of solar power as a basis for environmentally clean aviation.

SOLAR IMPULSE 2

This propeller-driven aircraft has over 17,000 solar cells on its upper surfaces powering four electric motors. Built of carbon fiber, it weighs little more than a large car, yet had almost the wingspan of an A380.

ALTERNATING PILOTS

Bertrand Piccard, shown here in China during the voyage, alternated with André Borschberg as pilot of the single-seat aircraft. The demands of flying solo for journeys lasting up to five days were extreme.